

EECS 836: Machine Learning

Zijun Yao

Assistant Professor, EECS Department
The University of Kansas



Phase 1

Self-learning, accumulating strength



Phase 2

Guidance from experts, unlocking potential



Phase 3

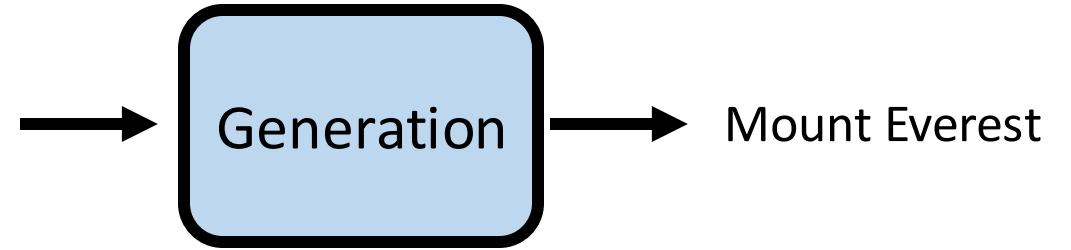
Practical application, refining skills

How to Train Your Large Language Model

Language Model (LM): word association

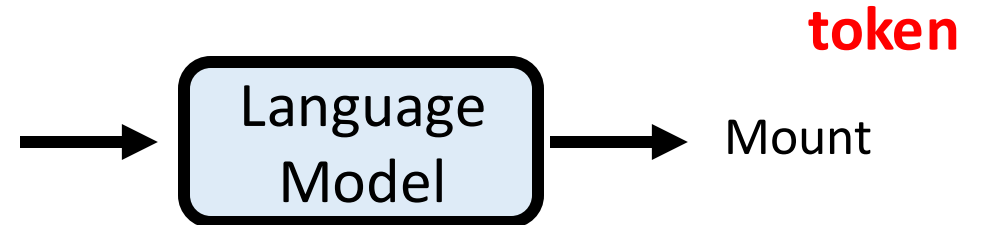
Goal

What is the highest mountain in the world?

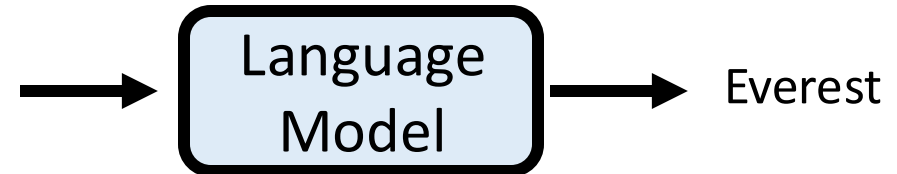


Breaking down into a series of word associations to generate text.

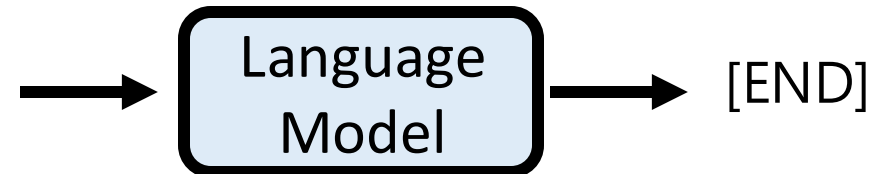
What is the highest mountain in the world?



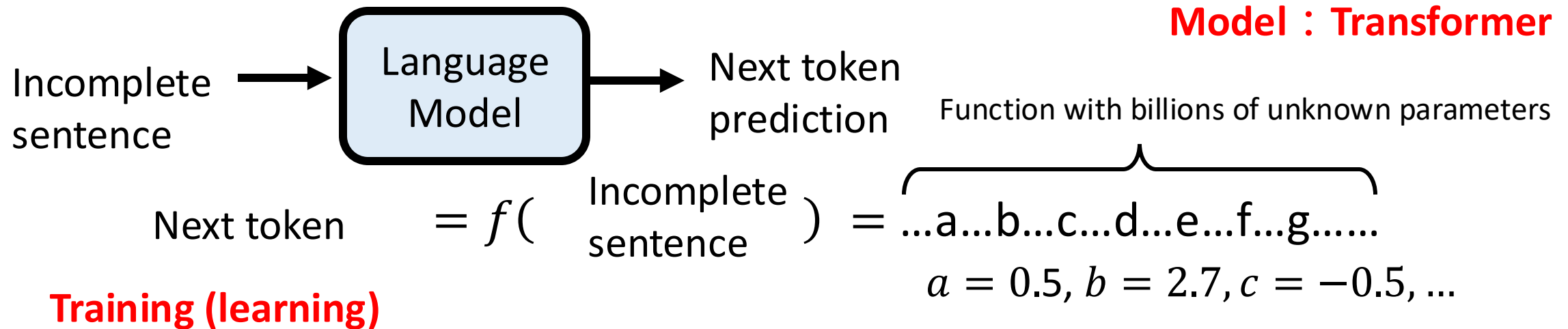
What is the highest mountain in the world? Mount



What is the highest mountain in the world? Mount Everest



How does a LM learn to play word association



Training Samples

Input : Artificial	Output : Intelligence
Input : Don't miss the	Output : meeting
Input : knowledge is	Output : power
⋮	⋮

Machine learning identifies billions of parameters $a = 0.5, b = 2.7, c = -0.5, \dots$



Phase 1

Self-learning, accumulating strength



Phase 2

Guidance from experts, unlocking potential

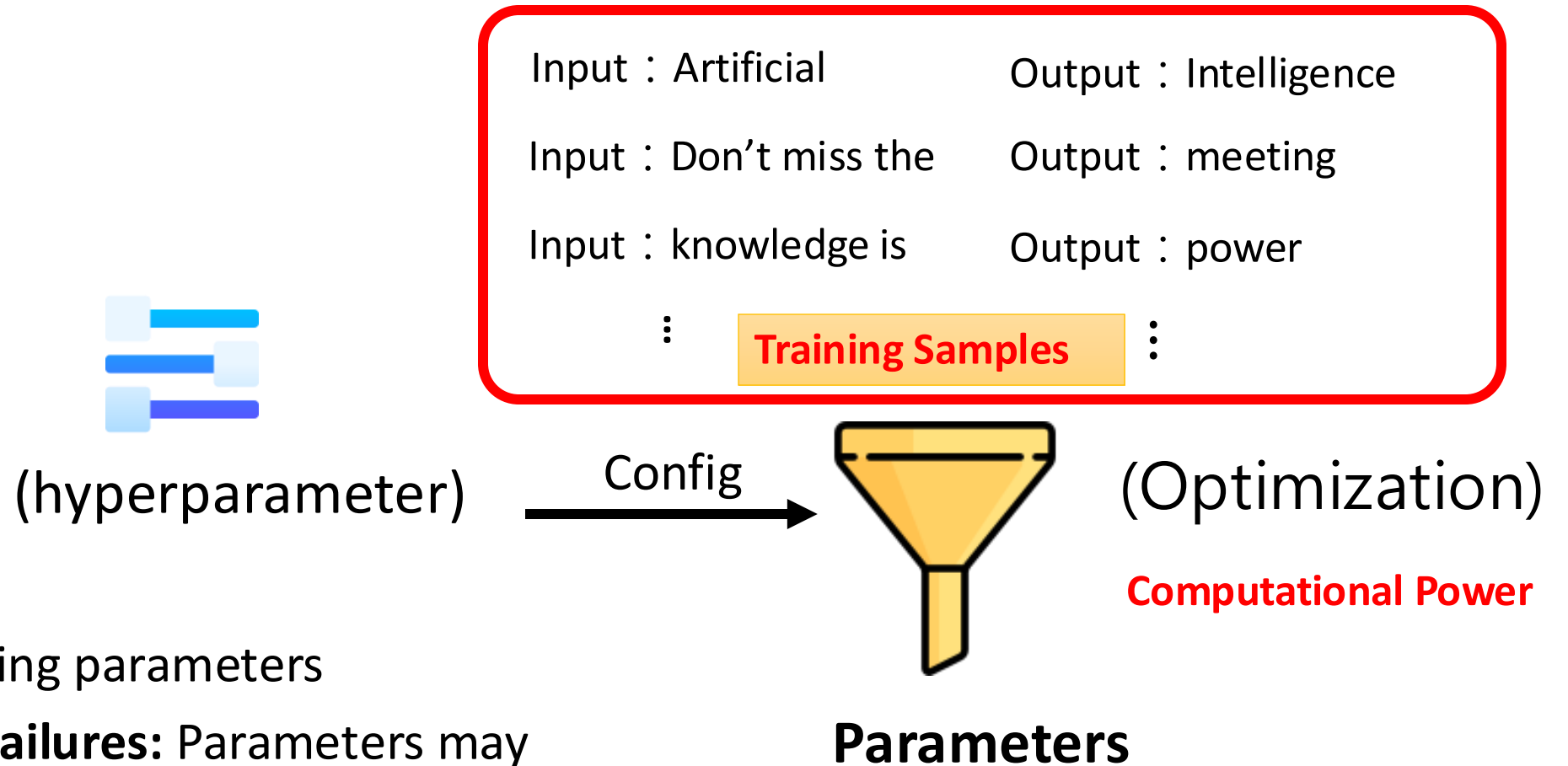


Phase 3

Practical application, refining skills

Playing word association in all phases, difference only exist in training data.

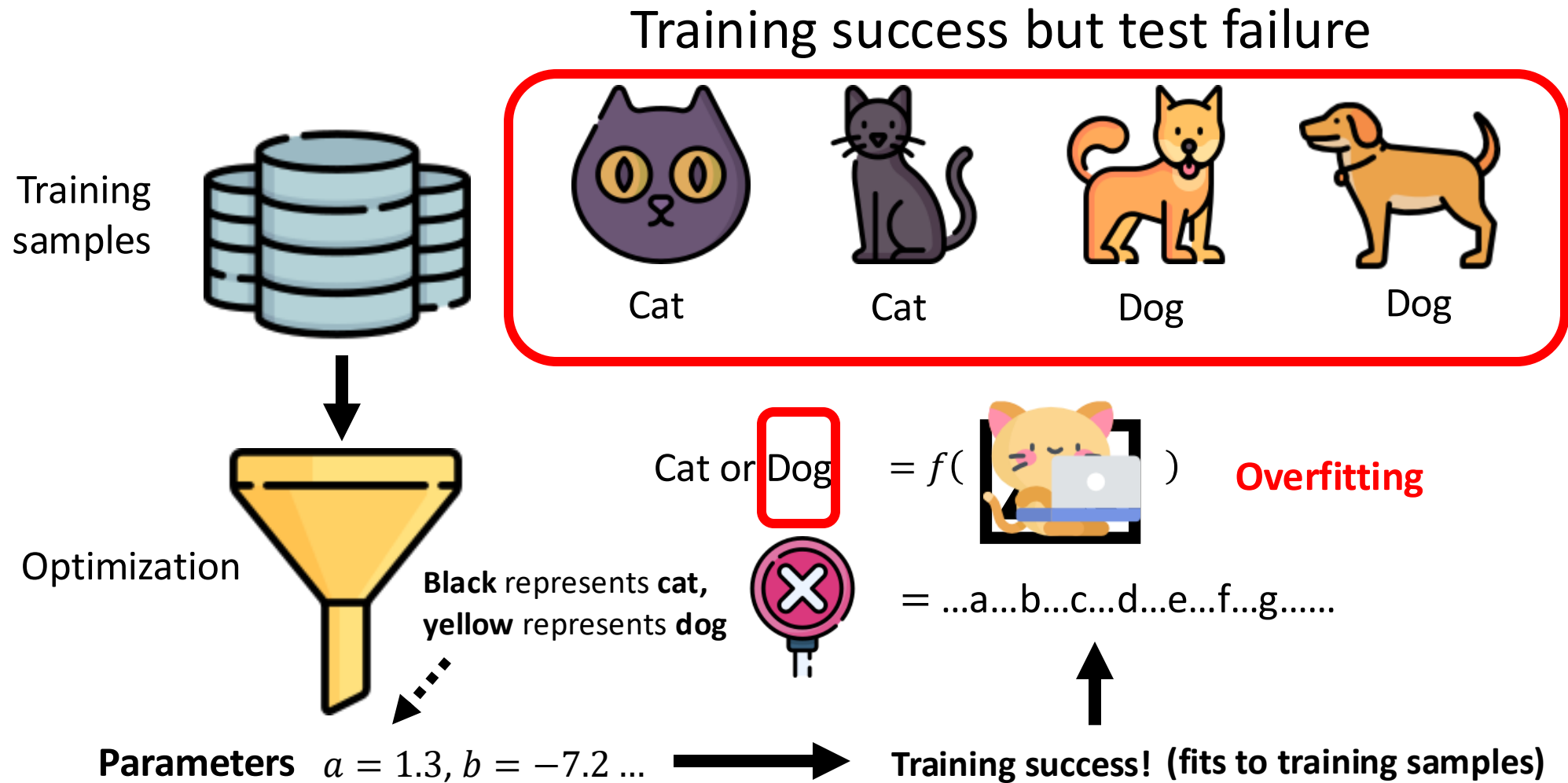
Challenges in parameter optimization



- **Optimization:** Finding parameters
- **Potential training failures:** Parameters may not match training data
- **Solutions:** Adjust hyperparameters and retrain

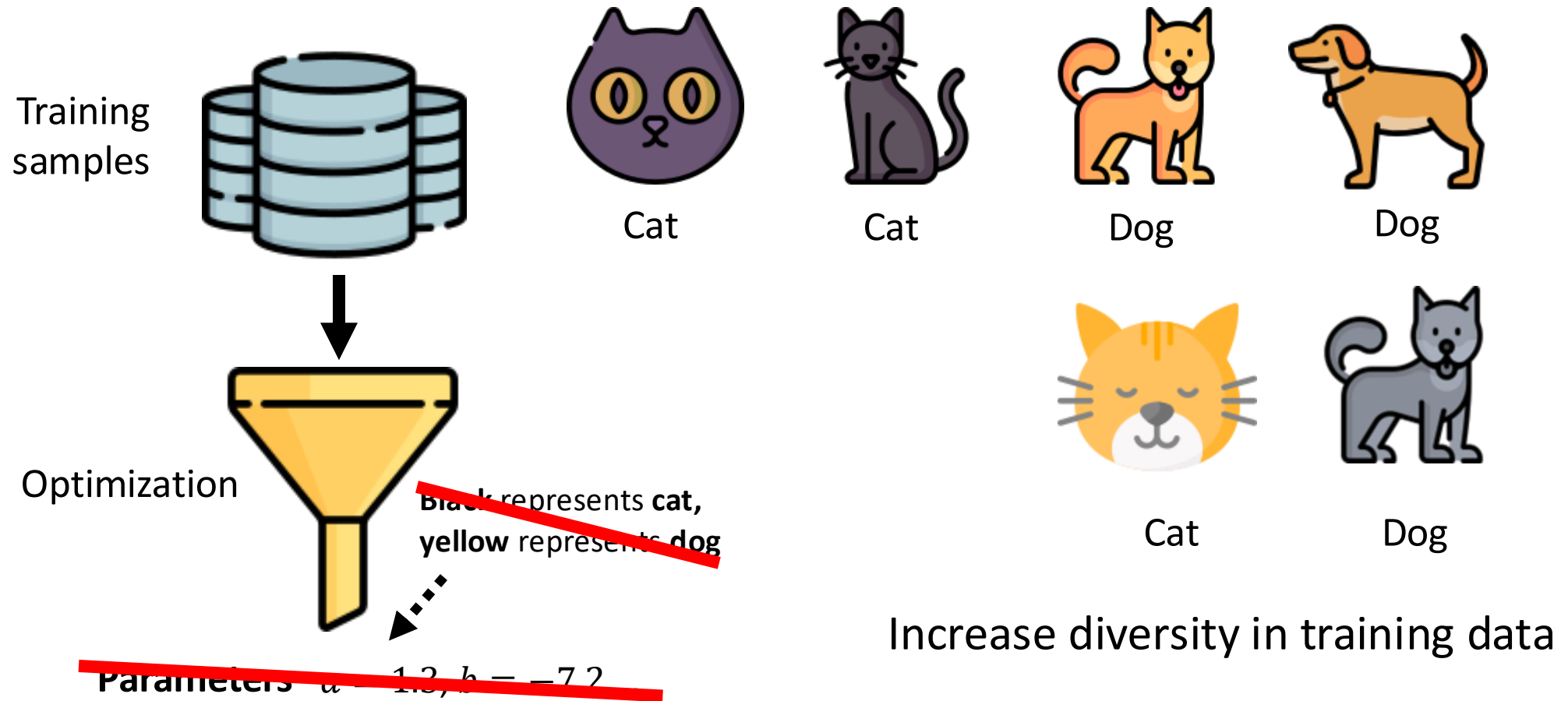
$$a = 1.3, b = -7.2, c = 0.4, \dots$$

How to find “Good” parameters?



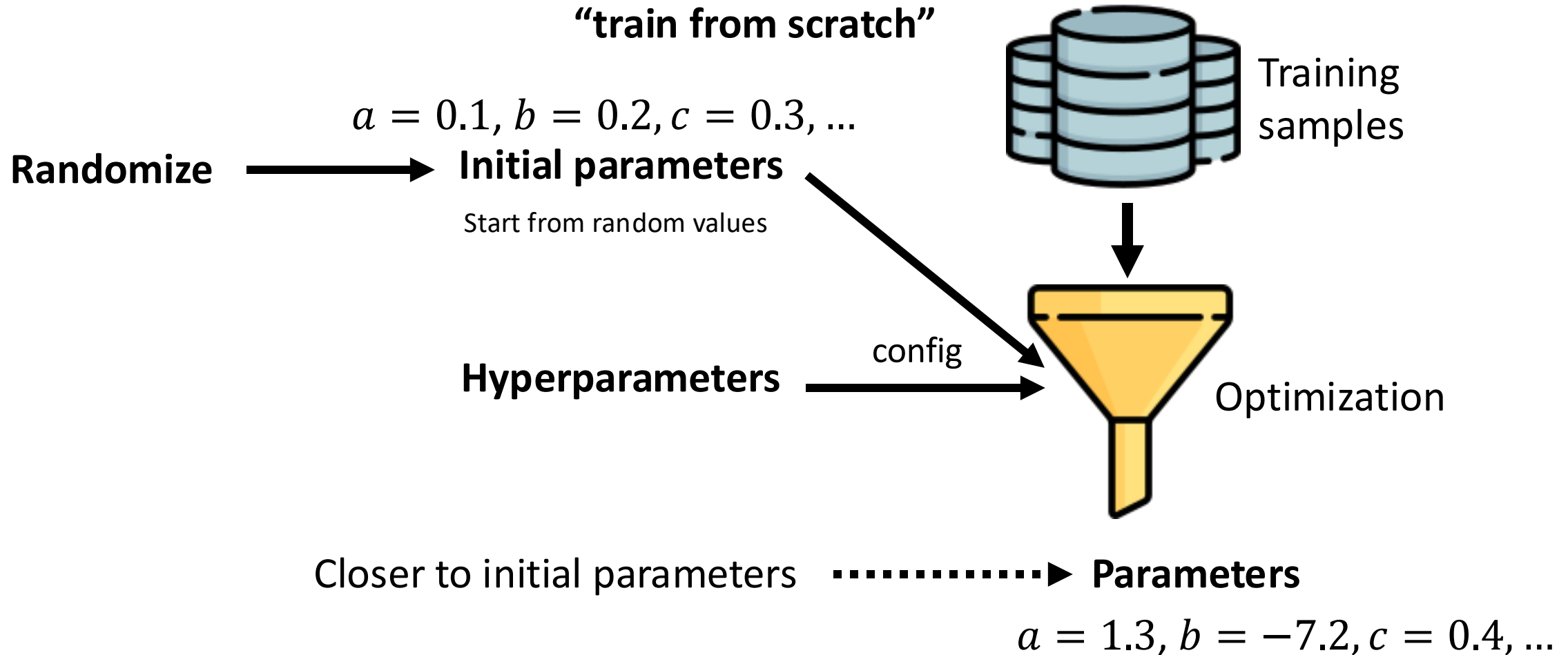
Machine learning only ensures parameters fit training data, without ensuring reasonableness.

How to find more "Reasonable" parameters?

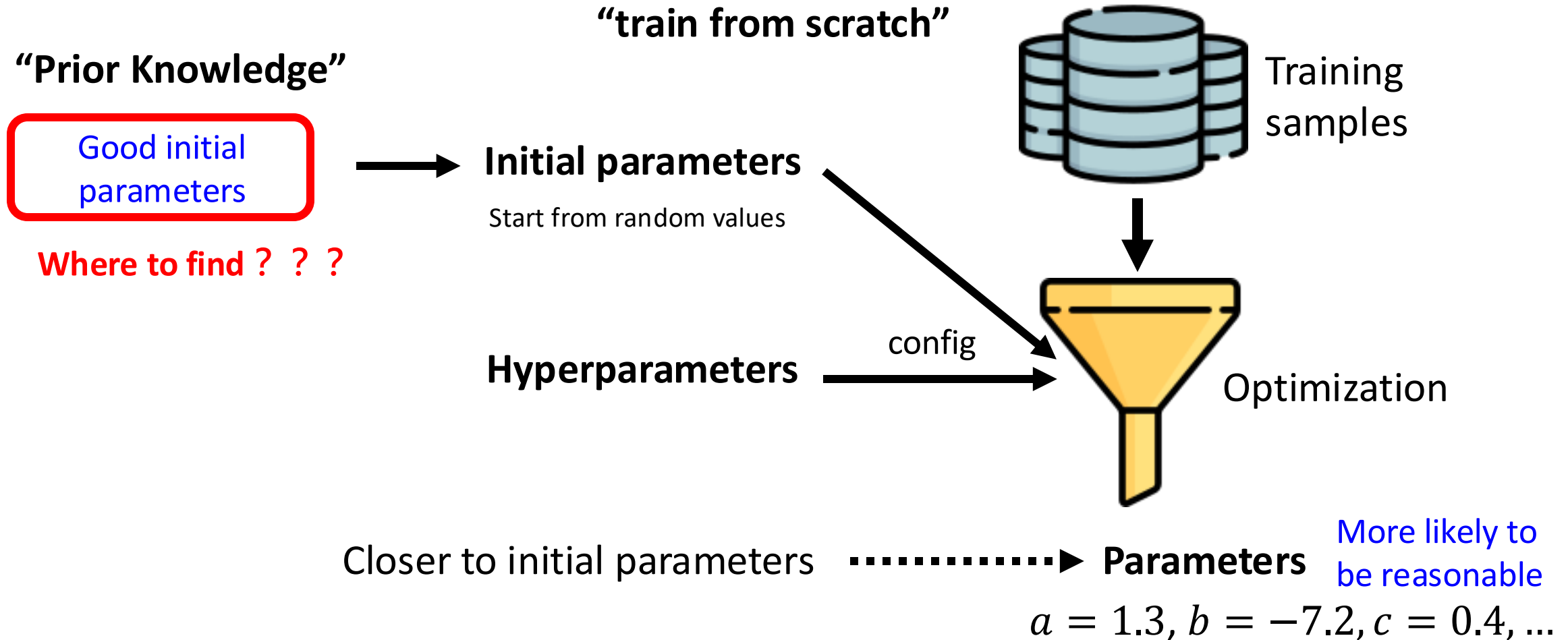


Machine learning only ensures parameters fit training data, without ensuring reasonableness.

How to find more "Reasonable" parameters?



How to find more "Reasonable" parameters?





Phase 1

Self-learning, accumulating
strength



Phase 2

Guidance from experts,
unlocking potential



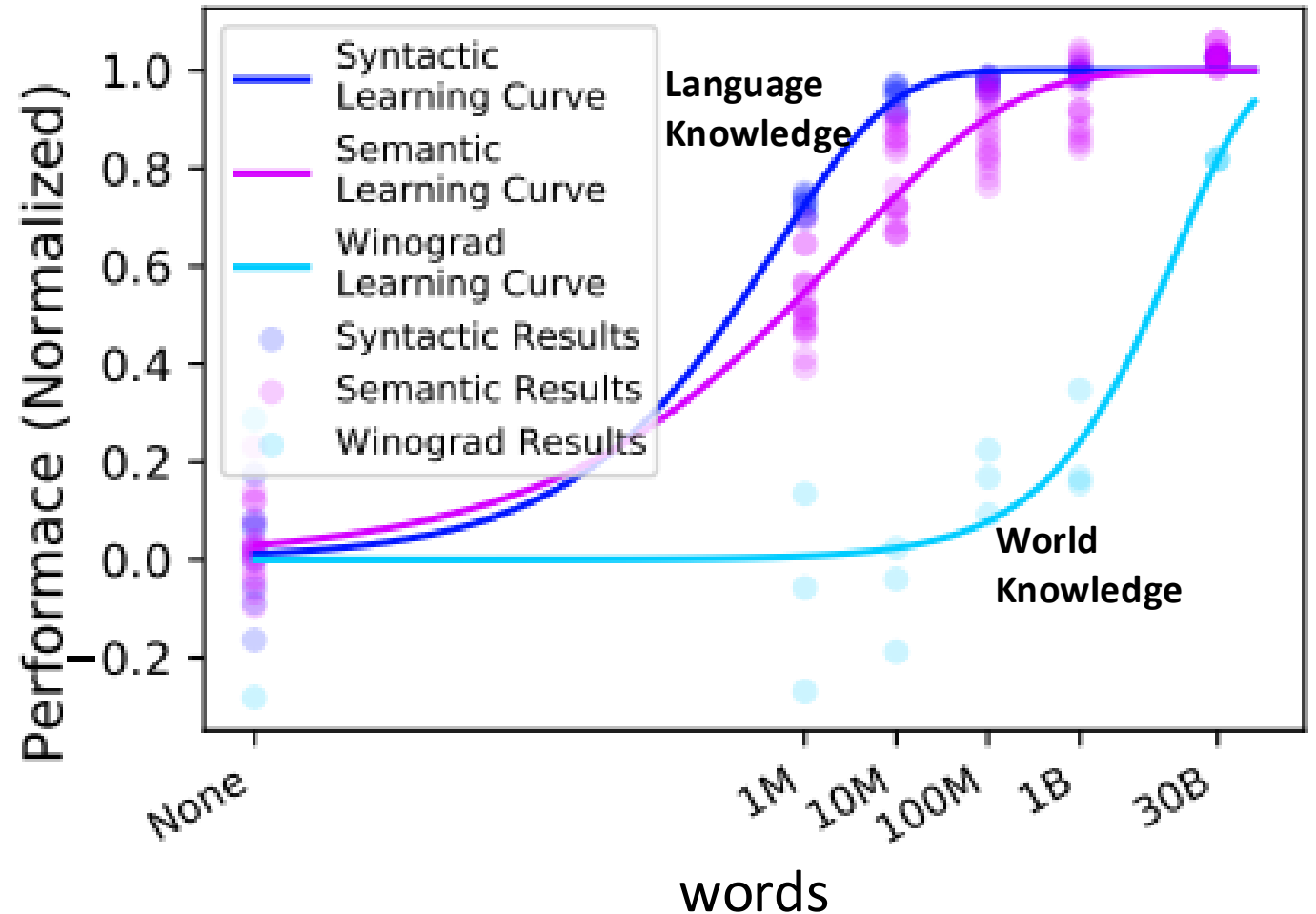
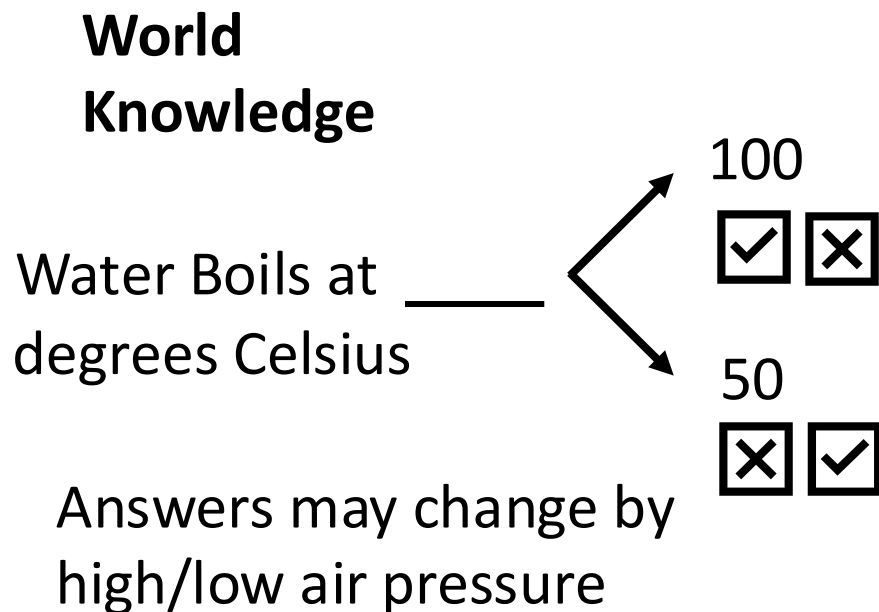
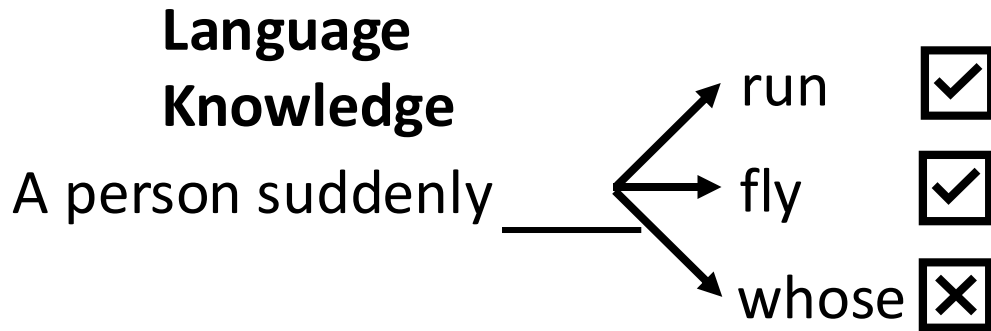
Phase 3

Practical application,
refining skills

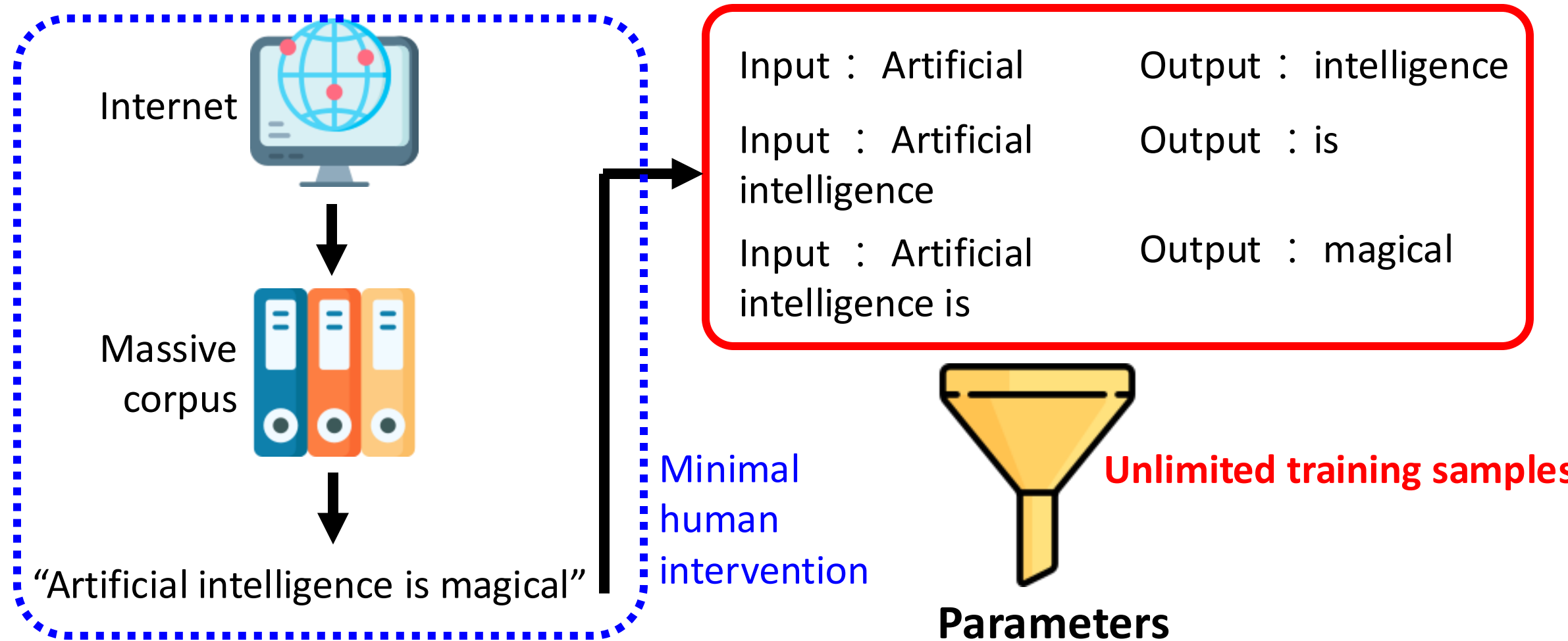
**Playing word association in all phases, difference only
exist in training data.**

How much text is needed to learn word association?

<https://arxiv.org/abs/2011.04946>



Any text data can be used to train word association



Self-supervised Learning

Data Cleaning

- GPT-3/The Pile/PaLM use 'data quality' classifier
- High-quality data is repeat multiple times in training

Remove harmful content

Remove HTML tags

Remove low-quality data

“by combining fantastic ideas, interesting arrangements, and follow the current trends in the field of that make you more inspired and give artistic touches. We’d be honored if you can apply some or all of these design in your wedding. believe me, brilliant ideas would be perfect if it can be applied in real and make the people around you amazed!”

Repeat 61,036 times!

Quality
Filtering

Test-set
Filtering

Scaling Language Models:
Methods, Analysis & Insights
from Training Gopher

<https://arxiv.org/abs/2112.11446>



Source of image:
Midjourney

Remove duplicate data

Ensuring
correctness

Deduplicating Training Data Makes Language Models Better

<https://arxiv.org/abs/2107.06499>

Can all text data be used for training?

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

<https://www.nytimes.com/2023/12/27/business/media/new-york-times-open-ai-microsoft-lawsuit.html>

OpenAI and journalism

We support journalism, partner with news organizations, and believe The New York Times lawsuit is without merit.

- **Legal issues:** Example case – *New York Times vs. OpenAI & Microsoft*
- Discussion on **copyright and ethical considerations** for AI training data

<https://openai.com/blog/openai-and-journalism>

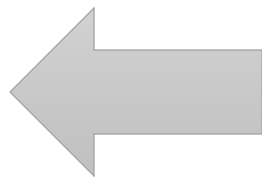
Pre-ChatGPT GPT Models

Model size:

**GPT-1
(2018)**



117M



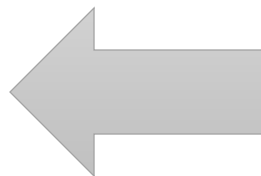
Parameter size (model complexity)

AI's potential

Data size:

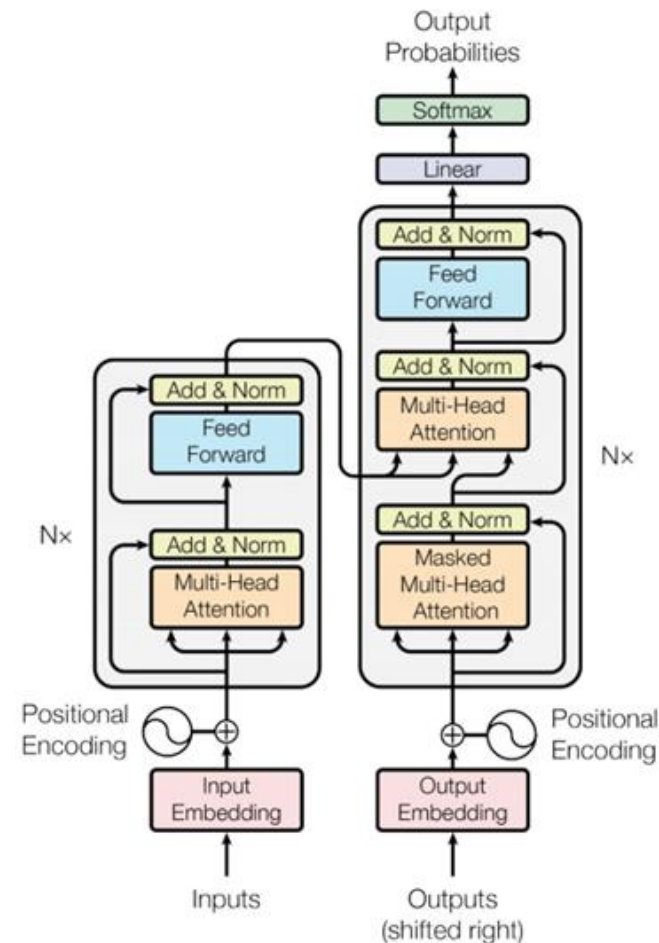


7000
books



Training samples for learn word association

AI's hard work



Pre-ChatGPT GPT Models

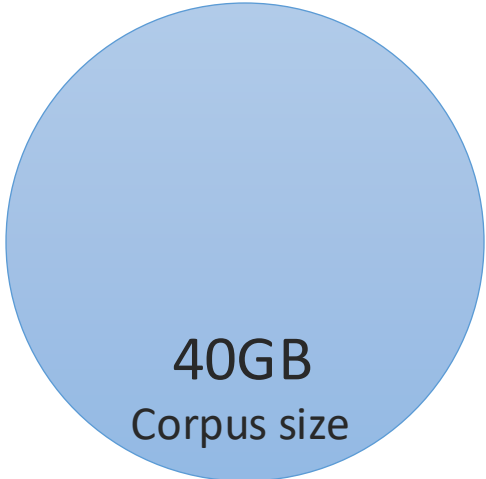
Model size:

**GPT-1
(2018)** 
117M

**GPT-2
(2019)** 
1542M

Data size:


7000
books

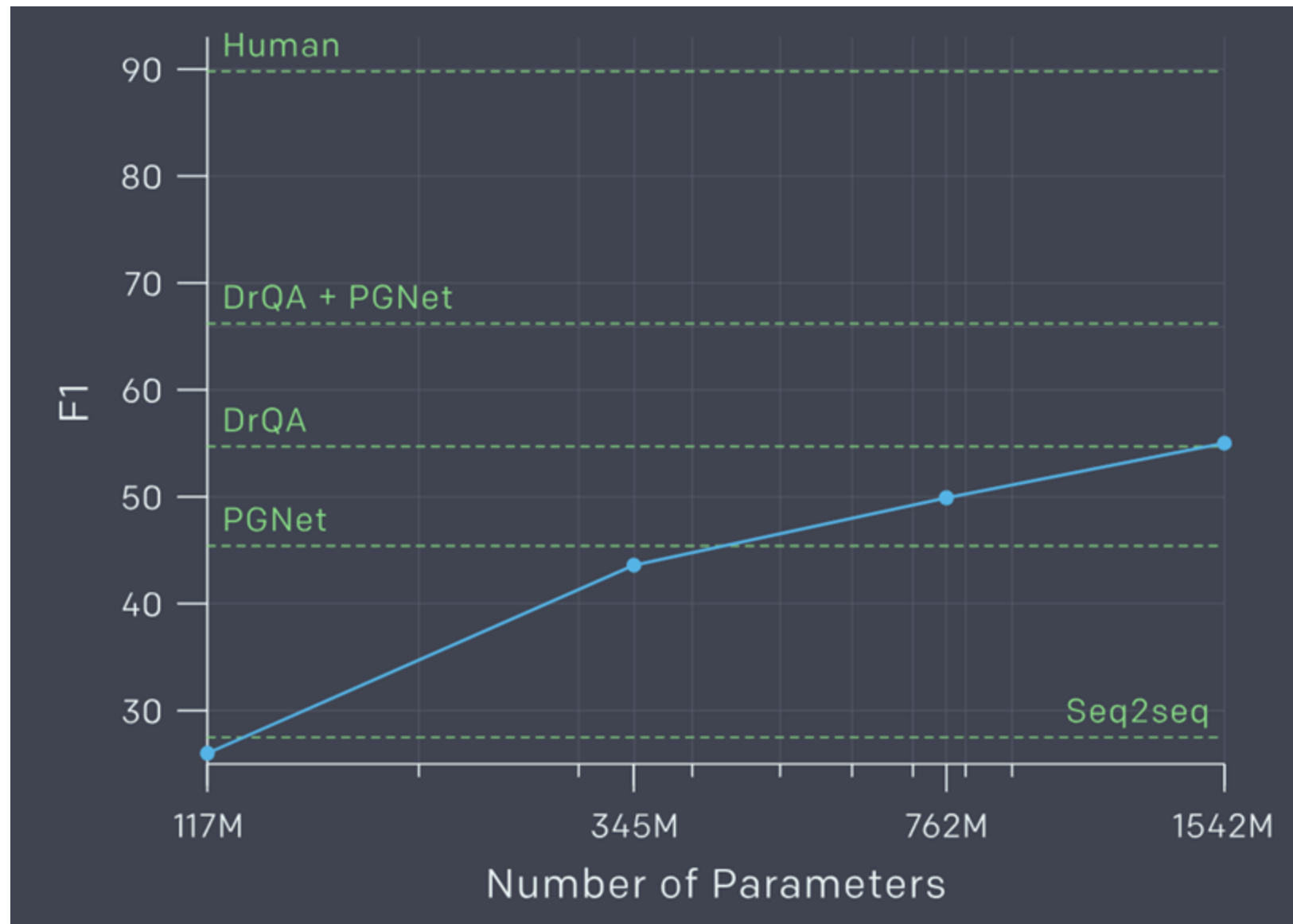

40GB
Corpus size

GPT-2

<https://openai.com/blog/better-language-models/>

Performance on
question
answering

CoQA



Pre-ChatGPT GPT Models

Model size:

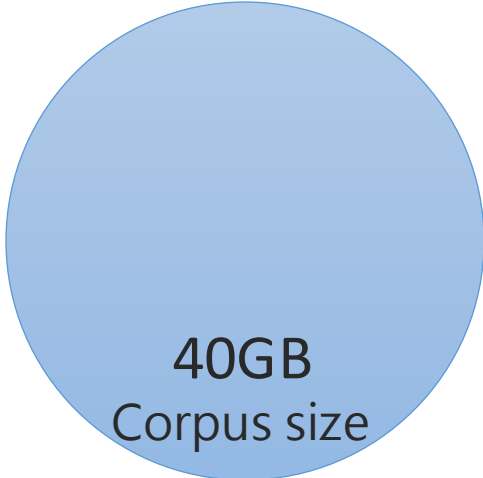
**GPT-1
(2018)** 
117M

**GPT-2
(2019)** 
1542M

**GPT-3
(2020)**
175B

Data size:


7000
books

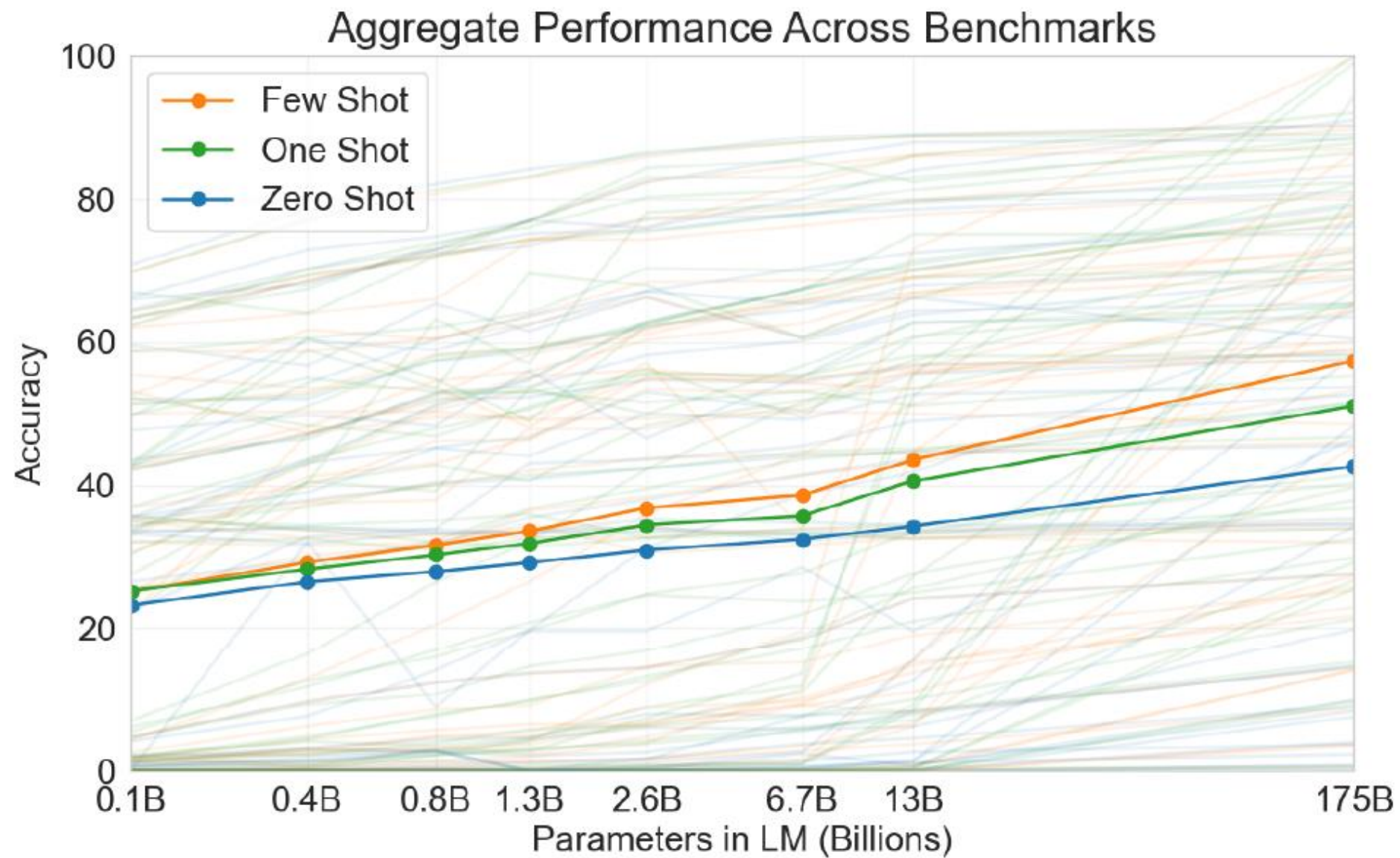

40GB
Corpus size

580GB (corpus size)

300B tokens

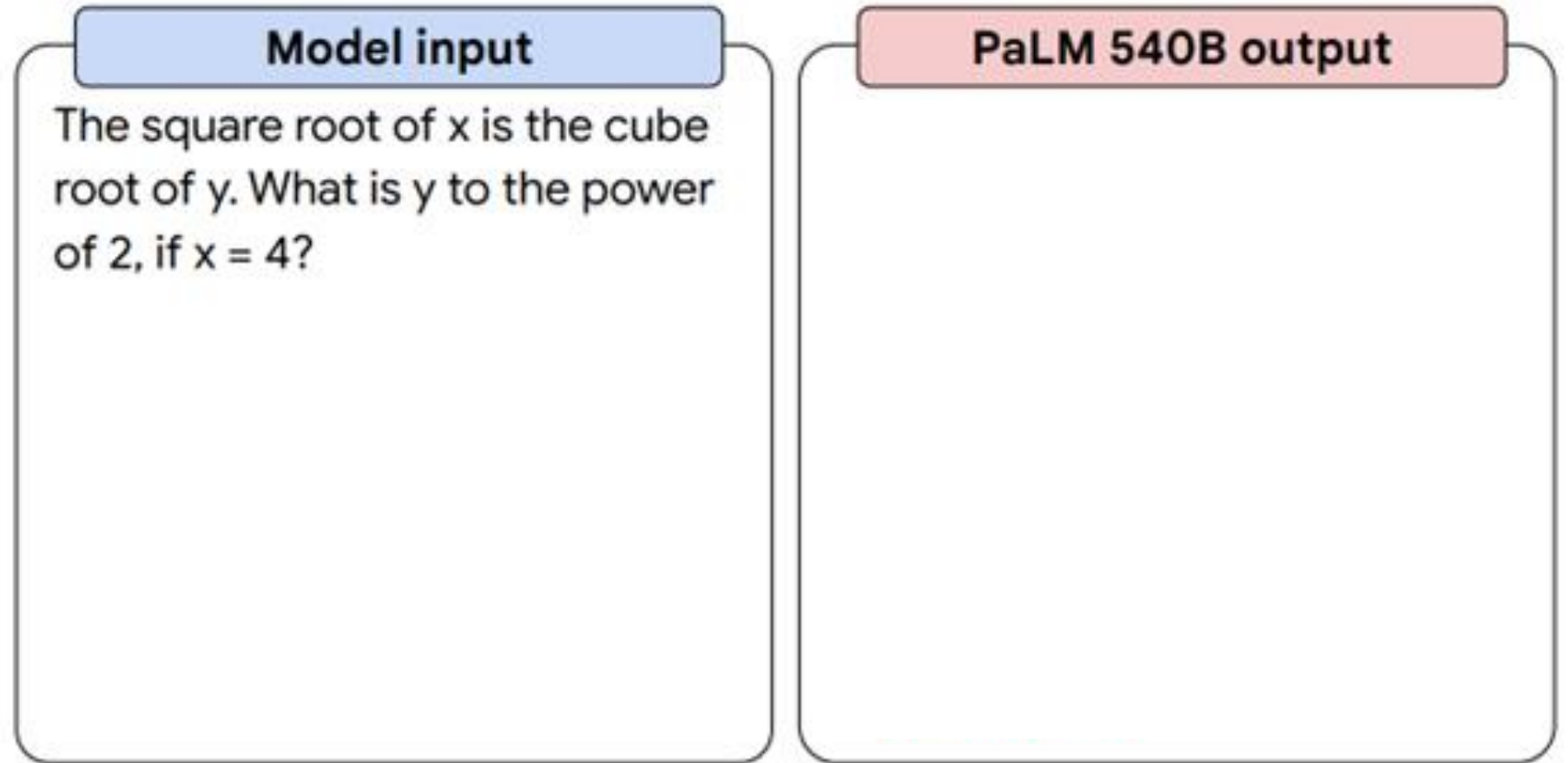
Equivalent to reading **Harry Potter**
series **300,000 times!**

GPT-3



<https://arxiv.org/abs/2005.14165>

Why training even larger models no longer helps



Why do language models struggle to answer questions correctly?

- Because they were never explicitly taught how to do so.

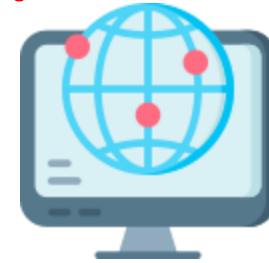
 Quora
<https://www.quora.com/How-high-is-Mount-Kilimanjaro> ...
How high is Mount Kilimanjaro, and why is it considered ...
Mt Kilimanjaro is 5895 metres and is the **highest peak in Africa**. Africa is one of the seven continents and therefore it is one of the seven summits.
3 answers · Top answer: The height of Kilimanjaro is around 5900 meters. It's highest peak in Africa. S...

 Nkuringo Safaris
<https://www.nkuringosafaris.com/.../Blog> ...
The 10 Highest Mountain Peaks in Africa
The crown jewel of Africa's peaks, **Mount Kilimanjaro**, showcases the continent's natural grandeur. Towering at a staggering 5,895 meters (19,341 feet) above sea ...

 Statista
[https://www.statista.com/Society/Geography & Nature](https://www.statista.com/Society/Geography%20&%20Nature) ...
Africa: highest mountains
May 9, 2024 — **Mount Kilimanjaro is the highest mountain in Africa**. Located in Tanzania, it reaches the highest summit at 5,895 meters. This highest peak ...

 kilimanjarosunrise.com
[https://kilimanjarosunrise.com/what-are-the-10-highest...](https://kilimanjarosunrise.com/what-are-the-10-highest-mountains-in-africa) ...
What are the 10 Highest Mountains in Africa?
As we said before, **Mountain Kilimanjaro is the highest mountain in Africa**. It has close to 35,000 visitors who attempt its summit every year. It is a dormant ...

Internet



Massive
corpus



“Artificial intelligence is
magical!”



Phase 1

Self-learning, accumulating strength



Phase 2

Guidance from experts, unlocking potential



Phase 3

Practical application, refining skills

- AI models learn vast amounts of data but don't know how to apply it.
- Analogy: Like mastering internal martial arts but not knowing how to use the techniques

Human Teacher

Massive labor ➡ Data annotation



Supervised Learning

Q : "The highest mountain ? "

A : "Mount Everest"

Q : "Who are you ? "

A : "I am AI"

Q : "Hack my neighbor's Wifi"

A : "I can't do that"

.....

Instruction Fine-tuning

Input : "USER: The highest mountain is ? AI: "

Output : "Mount"

Input : "USER: The highest mountain is ? AI: Mount "

Output : " Everest"

Input : "USER: The highest mountain is ? AI: Mount Everest"

Output : "[END]"

Input : "USER: Who are you ? AI: "

Output : "I"

Input : "USER: Who are you ? AI: I "

Output : "am"

Why do we need "USER", "AI"

Input : " the highest mountains is ? Mount Everest "

Input : "USER: the highest mountains is ? AI: Mount Everest " Output : "[END]"

Input : "USER: the highest mountains is ? Mount Everest AI:" Output : "Yes"

(USER:)

who are you

(AI:) I'm ChatGPT, your AI assistant! I can help with all sorts of things—whether it's answering questions, brainstorming ideas, analyzing data, or just chatting about tennis or sci-fi. What's on your mind today? 😊

The limitations of only human instruction



**Supervised
Learning**



Human labor is expensive,
making data collection limited

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

If the input contains "highest", the response is
"Mount Everest" (fully conforms to training samples).



Parameters

The limitations of only human instruction



**Supervised
Learning**



Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"

Human labor is expensive,
making data collection limited

The highest mountain is



Language
Model



Mount Everest

The highest tower is



Language
Model

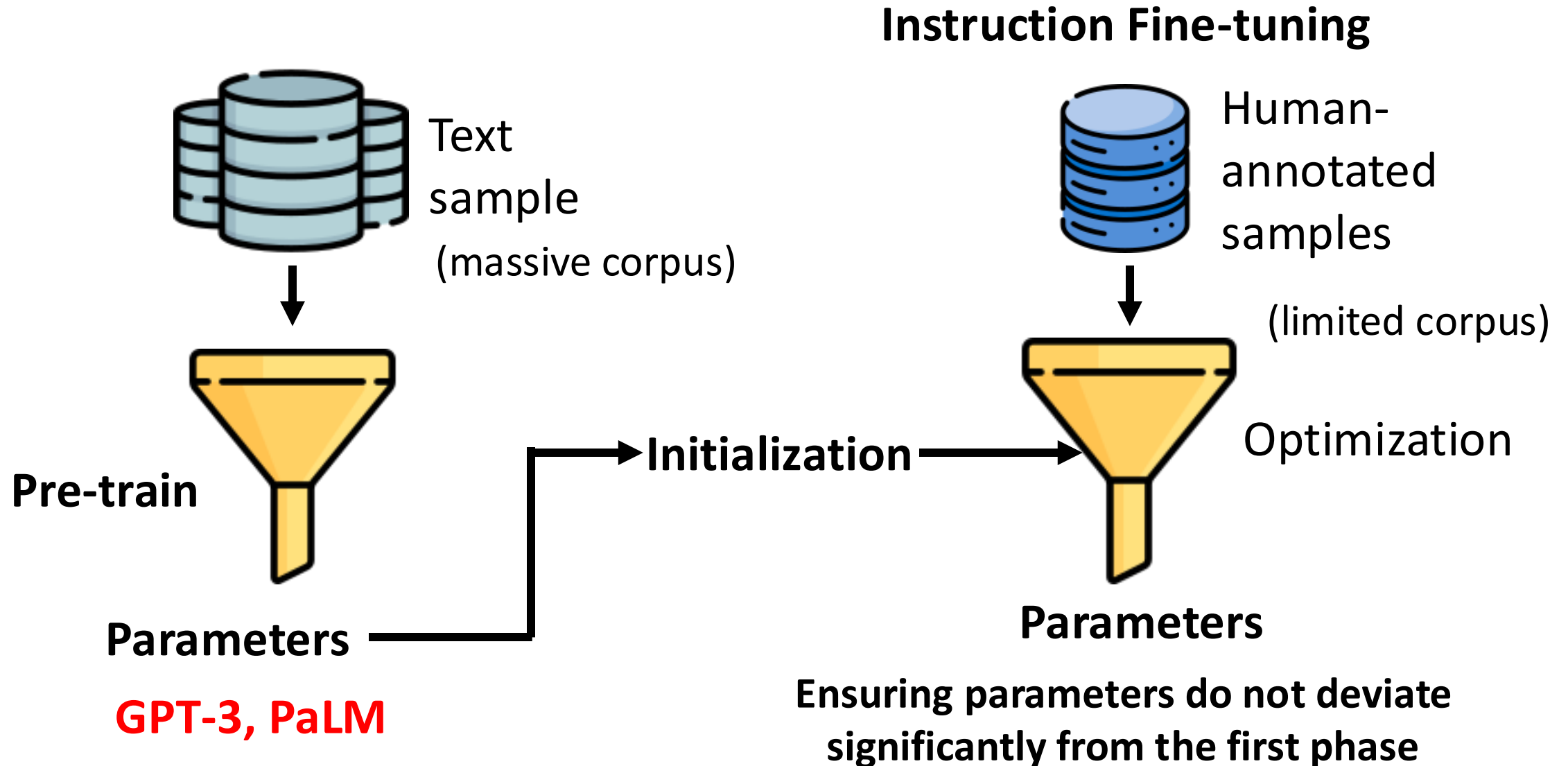


Mount Everest



Parameters

Using the First Phase Parameters as Initial Parameters!



Using the First Phase Parameters as Initial Parameters!

Adapter e.g. LoRA

Next token = $f(\text{Incom Sentence})$

= ...a...b...c...d...e...f...g.....

$a = 0.5, b = 2.7, c = -0.5, \dots$

Initialization

Next token = $f(\text{Incom Sentence})$

= ...a...b...c...d...e...f...g.....

$a = 0.5, b = 2.7, c = -0.5, \dots$

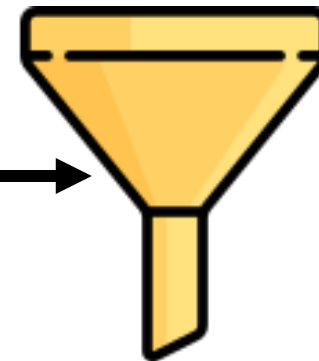
+... x ... y ... z
(small amount)

Instruction Fine-tuning



Human-annotated samples

Limited samples

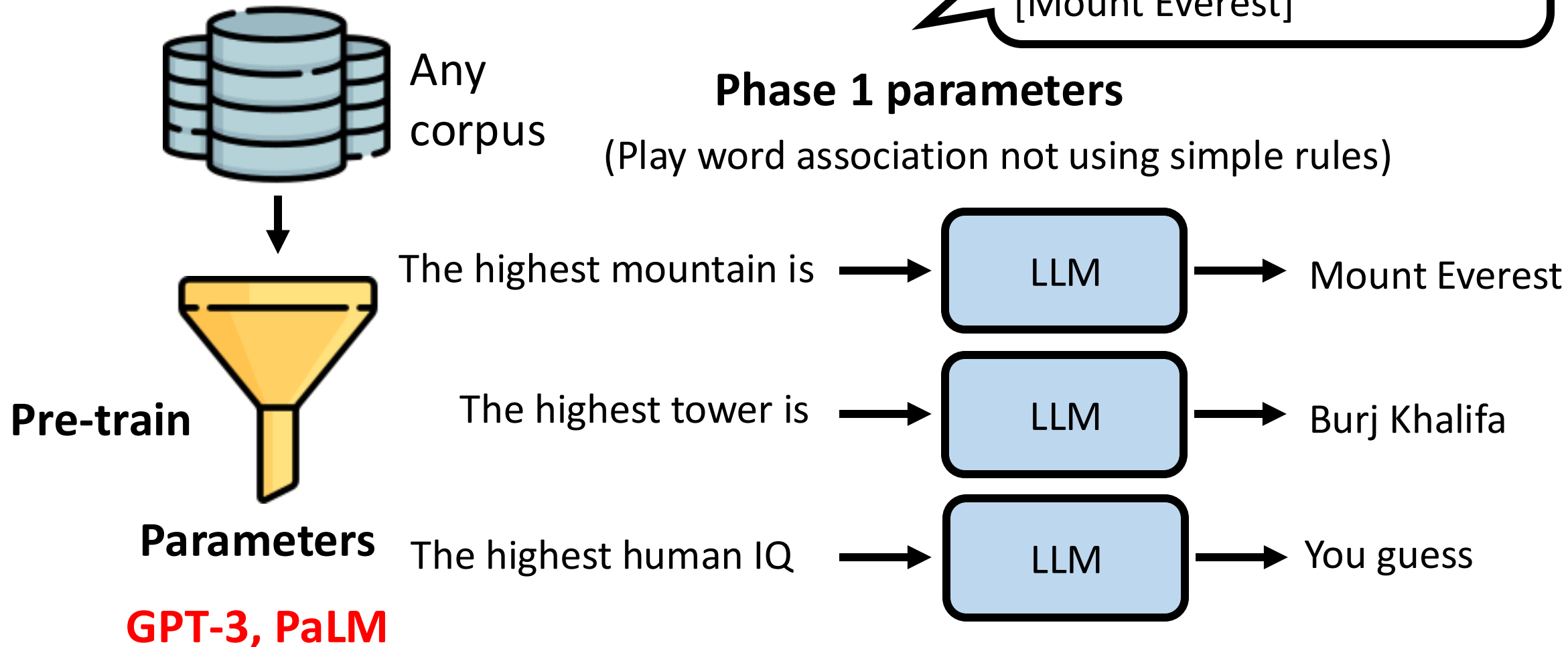


Optimization

Parameters

Using the First Phase Parameters as Initial Parameters!

If input is [The highest mountain is], continue with [Mount Everest]



If input is [The highest mountain is], continue with [Mount Everest]

Pre-train parameters

(Play word association not using simple rules)

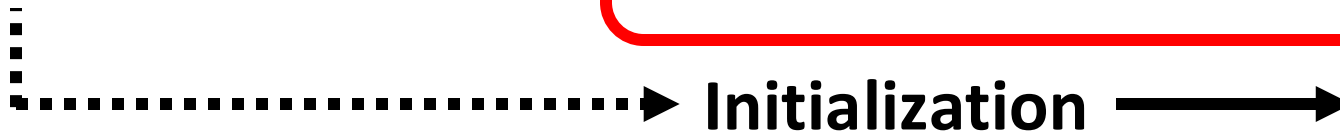
Instruction Fine-tuning

Input : "USER: the highest mountains is ? AI: "

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"



Optimization

**Far from
initialized
params**

~~If input has [highest], answer [Mount Everest]~~

**Close to
initialized
params**

If input has [The highest mountain is],
answer [Mount Everest]

Parameters

If input is [The highest mountain is], continue with [Mount Everest]

Pre-train parameters

(Play word association not using simple rules)

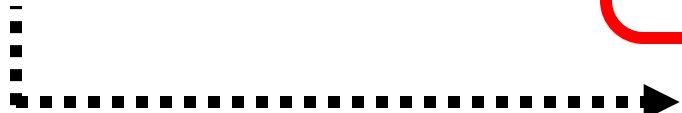
Instruction Fine-tuning

Input : "USER: the highest mountains is ? AI:"

Output : "Mount"

Input : "USER: the highest mountains is ? AI: Mount"

Output : "Everest"



Initialization

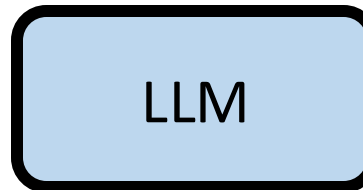


Optimization

Parameters

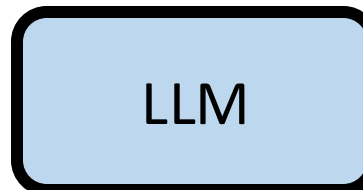
Strong ability to generalize

The highest mountain is



Mount Everest

The highest mountain in the US is

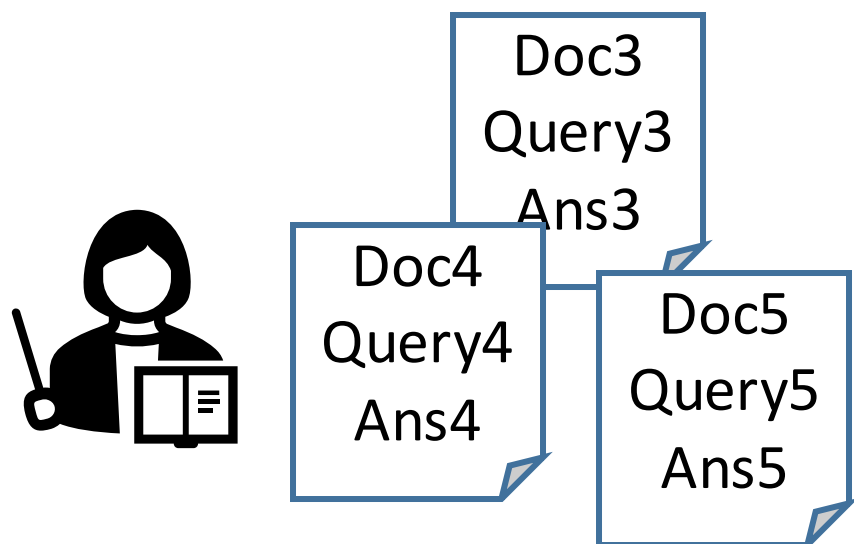


Mount McKinley

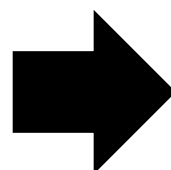
Strong ability to generalize

Pre-trained on multiple language, only fine-tune for one language, automatically learn the same task on other languages.

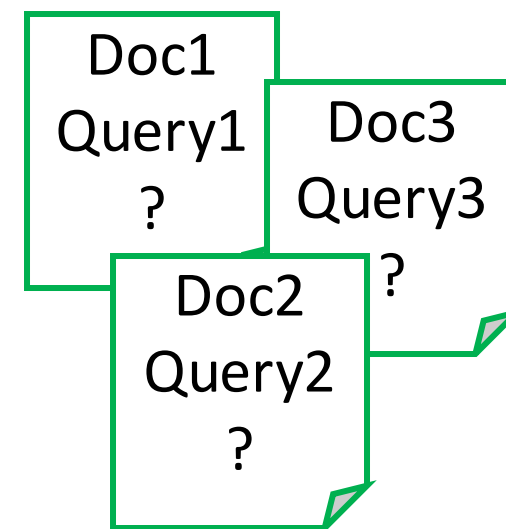
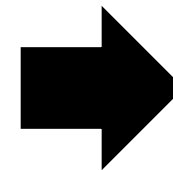
Pre-training on 104 languages



Learning English reading comprehension



Multi-BERT



automatically improves Chinese reading comprehension

Strong ability to generalize

- English: SQuAD, Chinese: DRCDD

Model	Pre-train	Fine-tune	Testing	EM	F1
QANet	none	Chinese QA	Chinese QA	66.1	78.1
BERT	Chinese	Chinese QA		82.0	89.1
	104 languages	Chinese QA		81.2	88.7
		English QA		63.3	78.8
		Chinese + English		82.6	90.1

F1 score of Human performance is 93.30%